

Hybrid Connectivity: Direct Connect and VPN

One in five questions on this exam has an on-premises data centre in it.

Direct Connect is private but **not encrypted**. VPN is encrypted but rides the **public internet**. Most real designs use both.

Direct Connect or VPN?

DIRECT CONNECT

A private circuit

- 1, 10, 100 or 400 Gbps ports
- Consistent latency, no internet
- Weeks to months to provision
- Not encrypted by default

SITE-TO-SITE VPN

Encrypted over the internet

- 1.25 Gbps per tunnel, 5 with LBT
- Latency varies with the internet
- Up today, in minutes
- IPsec encrypted by default

THE TIEBREAKER

Urgency favours VPN. Sustained throughput and predictability favour Direct Connect. Most answers pair them.

The three virtual interface types

VIF TYPE	CONNECTS TO	USE IT FOR
Private	A VGW, or a DX gateway	Reaching VPCs over private addresses
Public	AWS public endpoints	S3, DynamoDB and public services over the circuit
Transit	A DX gateway only	Reaching a Transit Gateway, and everything behind it

ON THE EXAM

Transit VIF is the only way to reach a Transit Gateway. If a scenario has a TGW and a Direct Connect, it needs one.

Direct Connect gateway

- ◆ **A global object.** One circuit reaches VPCs in any Region, except China.
- ◆ **20 virtual private gateways or 6 Transit Gateways** per DX gateway. Not both at once.
- ◆ It also removes the account problem: associate VGWs from many accounts with one gateway.
- ◆ **VGWs attached to a DX gateway cannot route to each other.** It is a hub for reaching AWS, not for VPC-to-VPC.

ON THE EXAM

A DX gateway does not give you transitive VPC connectivity. If VPCs must reach each other, you still need a Transit Gateway.

Follow the packet from the data centre

01

Customer router

Your router at the Direct Connect location, speaking BGP.

02

VIF

A VLAN on the circuit. Its type decides what is reachable.

03

DX gateway

Global. Routes on to a VGW or a Transit Gateway.

04

TGW route table

Module six's association decides where it goes next.

05

Subnet route

The destination VPC's own route table delivers it.

ON THE EXAM

Five hops, five places a route can be missing. Troubleshooting questions are almost always asking which one.

The four resiliency models

MODEL	WHAT IT IS	FITS
Maximum	Separate connections at two or more locations, separate devices	Critical production
High	One connection at each of two locations	Most production
Development and Test	Two connections, separate devices, one location	Non-critical only
Classic	Added without the toolkit, no resiliency	Existing single links

ON THE EXAM

The word to hunt for is location. Two connections in one building survive a device failure and not a building failure.

Circuit-level details worth knowing

- ◆ **A hosted connection supports exactly one VIF.** A dedicated connection supports 51. This decides scenarios.
- ◆ **LAG** bundles connections into one logical link. Four below 100G, two at 100G.
- ◆ **MACsec** encrypts at layer 2 on supported dedicated ports, between your router and the AWS router only.
- ◆ **SiteLink** sends traffic between two Direct Connect locations over the AWS backbone, with no VPC in the path.

ON THE EXAM

MACsec protects the circuit, not the journey. If a partner sits in the path, it does not cover that segment.

VPN numbers that decide answers

- ◆ **Every connection has two tunnels.** Standard is 1.25 Gbps each; Large Bandwidth Tunnels reach 5 Gbps.
- ◆ **Large Bandwidth Tunnels need a Transit Gateway.** They do not work on a virtual private gateway.
- ◆ **A VGW caps at about 1.25 Gbps in aggregate,** however many VPN connections you attach to it.
- ◆ **ECMP** aggregates tunnels for more bandwidth. It needs BGP, and it also needs a Transit Gateway.

ON THE EXAM

Every route past the VPN bandwidth ceiling runs through a Transit Gateway. A VGW is the ceiling itself.

Encrypting traffic over Direct Connect

MACSEC

Layer 2 on the circuit

- Supported dedicated ports only
- Line rate, no throughput penalty
- Your router to the AWS router
- Does not cover a partner segment

PRIVATE IP VPN

IPsec inside the circuit

- Runs over a transit VIF
- Subject to VPN tunnel bandwidth
- End to end, whoever is in the path
- Needs a Transit Gateway

THE TIEBREAKER

Line-rate encryption on your own circuit means MACsec. Encryption end to end, or over a hosted connection, means a VPN.

Steering traffic with BGP

- ◆ **Longest prefix match wins first**, before anything else. Advertise a more specific route to pull traffic.
- ◆ **AS_PATH prepending** makes a path look longer, so AWS prefers the other one. The usual way to mark a backup.
- ◆ **Local preference BGP communities** let you set preference on AWS's side of the connection.
- ◆ **Direct Connect is preferred over VPN** to the same destination by default, so failover is automatic.

ON THE EXAM

Traffic taking the wrong circuit is almost always a prefix that is more specific somewhere it should not be.

Reading the scenario

THE SCENARIO SAYS

Connectivity needed this week

Direct Connect must reach a Transit Gateway

Survive a Direct Connect location outage

Regulator requires encryption over the circuit

More VPN throughput than 1.25 Gbps

Traffic prefers the wrong circuit

IT IS POINTING AT

Site-to-Site VPN, possibly then DX

Transit VIF plus a DX gateway

Maximum or High resiliency

MACsec, or private IP VPN

Transit Gateway, then LBT or ECMP

A more specific prefix somewhere

ON THE EXAM

Check what terminates the connection first. A virtual private gateway is the constraint behind several of these answers.

Meridian Freight has one 10 Gbps Direct Connect at a single location, terminating on a virtual private gateway. A recent outage at that Direct Connect location took the data centre offline from AWS for six hours.

The business now requires that no single facility failure can sever connectivity, and wants an encrypted fallback path.

What should the solutions architect recommend?

- ☐ A Add a second connection on a different device at the same location
- ☐ B Add a connection at a second Direct Connect location, and a VPN as backup
- ☐ C Increase the existing connection to 100 Gbps
- ☐ D Add a LAG bundling two connections at the existing location

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Second device, same location

This is the Development and Test model. It survives a device failure and not the facility failure that actually happened.

C Upgrade to 100 Gbps

Answers a bandwidth question nobody asked. One faster circuit in one building is still one building.

B Second location plus VPN

Two locations meet the facility requirement, and the VPN adds the encrypted fallback. DX is preferred over VPN by default, so failover is automatic.

D LAG at one location

A LAG is one logical link for capacity and device redundancy. Every member terminates at the same location.

THE SIGNAL

No single facility failure eliminates every same-location answer at once. Three of the four options are in one building.

A company has a Direct Connect with a private VIF to a virtual private gateway in one VPC. It has since built a Transit Gateway with 40 VPCs attached, and wants the data centre to reach all of them.

An engineer proposes creating 40 private VIFs, one per VPC.

What should the solutions architect recommend?

- ☐ A Create the 40 private VIFs as proposed
- ☐ B Create a transit VIF to a Direct Connect gateway, associated with the Transit Gateway
- ☐ C Associate the existing virtual private gateway with the Transit Gateway
- ☐ D Peer the 40 VPCs with the VPC that holds the virtual private gateway

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** 40 private VIFs
Within the 50 VIF limit on a dedicated connection, and forty BGP sessions to operate. It also ignores the Transit Gateway that already exists.
- B** Transit VIF and DX gateway
One VIF, one BGP session, and every VPC behind the Transit Gateway becomes reachable, including ones added later.
- C** Associate the VGW
A DX gateway holds virtual private gateways or Transit Gateways, never both. A VGW cannot be associated with a Transit Gateway.
- D** Peer 40 VPCs
Peering is not transitive, from module six. On-premises traffic cannot pass through the peered VPC to reach the others.

THE SIGNAL

A transit VIF is the only route from Direct Connect to a Transit Gateway. If both appear in a question, that is the connection.

A bank buys a 1 Gbps hosted connection from a Direct Connect partner. Regulation requires all traffic between the data centre and AWS to be encrypted in transit, end to end.

The network team has asked whether MACsec on the connection satisfies the requirement.

What should the solutions architect advise?

- ☐ A Yes, enable MACsec on the hosted connection
- ☐ B No; run a private IP VPN over a transit VIF, so IPsec covers the whole path
- ☐ C No; encryption over Direct Connect is not possible, so use a Site-to-Site VPN over the internet
- ☐ D Yes, provided the application also uses TLS

◆ PAUSE AND DECIDE

What should the solutions architect advise?

A Enable MACsec

MACsec runs on supported dedicated ports, not hosted connections, and covers only router to router. The partner segment stays unprotected.

C Encryption over DX impossible

Simply false. Both MACsec and private IP VPN exist for this, and abandoning the circuit gives up its predictability for nothing.

B Private IP VPN over transit VIF

IPsec inside the circuit protects the traffic regardless of who is in the path, which is what end to end requires.

D MACsec plus TLS

TLS protects application traffic, and the requirement covers all traffic. It also does not fix MACsec's unavailability here.

THE SIGNAL

Hosted connection plus end to end is the pair that decides it. A partner in the path is exactly what MACsec does not cover.

A company runs four Site-to-Site VPN connections from its data centre to a virtual private gateway, expecting roughly 5 Gbps in total. Measured throughput from AWS to on-premises sits stubbornly near 1.2 Gbps regardless of how many connections are added.

All connections use BGP and are healthy.

What is the cause, and what should change?

- ☐ A A virtual private gateway has an aggregate throughput limit; terminate on a Transit Gateway instead
- ☐ B Enable ECMP on the virtual private gateway
- ☐ C Switch the connections to static routing to reduce BGP overhead
- ☐ D Request a quota increase for VPN tunnel bandwidth

◆ PAUSE AND DECIDE

What is the cause, and what should change?

- A

VGW aggregate limit; move to TGW
A VGW caps at roughly 1.25 Gbps in aggregate however many connections attach. Only a Transit Gateway lifts that ceiling.
- B

ECMP on the VGW
The closest miss. ECMP is real and is the right idea, and it is supported only on a Transit Gateway, never on a VGW.
- C

Static routing
Backwards. Static routing would remove ECMP as an option entirely, since it requires BGP.
- D

Quota increase
The aggregate limit is service behaviour rather than an adjustable quota. Support cannot raise it.

THE SIGNAL

Adding connections changing nothing is the giveaway. A per-connection problem would improve with more connections; a ceiling does not.

A company has Direct Connect at two locations, intending one as primary and one as backup. Both advertise the same 10.10.0.0/16 prefix, except that the backup site also advertises 10.10.5.0/24 for a legacy application.

Traffic destined for 10.10.5.0/24 always takes the backup circuit, even when the primary is healthy.

Why, and what should be changed?

- ☐ A AS_PATH prepending on the backup is missing; add it to fix the /24
- ☐ B Longest prefix match wins before any BGP attribute; advertise the /24 from both sites
- ☐ C The primary connection has a lower bandwidth and is being deprioritised
- ☐ D Direct Connect always load balances across all available circuits

◆ PAUSE AND DECIDE

Why, and what should be changed?

- A

 Add AS_PATH prepending
Prepending influences choice between routes of equal specificity. A /24 beats a /16 before AS_PATH is ever consulted.
- B

 Longest prefix wins; advertise from both
Specificity is decided first. Once both sites advertise the /24, the usual attributes take over and the primary is preferred.
- C

 Bandwidth deprioritisation
BGP does not select paths by circuit bandwidth. Nothing in the scenario mentions differing speeds anyway.
- D

 Always load balances
Not true, and it would not explain traffic consistently choosing one circuit.

THE SIGNAL

Traffic ignoring your carefully tuned preferences is nearly always specificity. Nothing outranks longest prefix match.

A retailer must connect a new distribution centre to AWS. The site opens in three weeks. Steady-state traffic is expected to be around 400 Mbps, rising sharply during seasonal peaks. A Direct Connect order has been placed and the partner quotes ten to twelve weeks.

What should the solutions architect recommend?

- ☐ A Delay the site opening until the Direct Connect is delivered
- ☐ B Establish a Site-to-Site VPN to the Transit Gateway now, and keep it as backup once DX is live
- ☐ C Cancel the Direct Connect order and rely on VPN permanently
- ☐ D Route the site's traffic over the existing data centre's Direct Connect

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A** Delay the opening
Subordinates a business event to a network order when a same-week option exists. Architecture rarely gets to set the opening date.
- B** VPN now, backup later
Up in minutes, comfortably above 400 Mbps, and it becomes the encrypted backup path once the circuit arrives.
- C** Cancel the DX order
Tempting given the numbers, and the scenario warns of sharp seasonal peaks that a single tunnel may not absorb.
- D** Backhaul via the data centre
Adds a hop and a dependency, and makes the existing data centre a single point of failure for a site that does not need one.

THE SIGNAL

Pair this with scenario one. There the VPN was the backup; here it is the bridge. Same service, different job.

What to carry out of this module

- ◆ **A transit VIF is the only path from Direct Connect to a Transit Gateway.** Both in a question means that VIF.
- ◆ **A hosted connection supports one VIF.** A dedicated connection supports 51.
- ◆ **A VGW caps near 1.25 Gbps in aggregate.** ECMP and 5 Gbps tunnels both need a Transit Gateway.
- ◆ **Longest prefix match beats every BGP attribute.** Wrong-circuit traffic is nearly always specificity.
- ◆ **MACsec protects the circuit.** A private IP VPN protects the journey, partners included.

ON THE EXAM

Check what terminates the connection before anything else. A virtual private gateway is the hidden ceiling behind several answers here.